

The resultant of these 2 E-fields would point upwards and to the left

23 Ans: B

Taking moments about axis of rotation,

Anti-clockwise moment due to magnetic force = clockwise moment due to mass

$$35 \times BIL \times 0.06 \times 2 = 0.20 (9.81) \times 0.06$$

$$35 \times (0.050 \times I \times 0.12) \times 0.06 \times 2 = 0.20 (9.81) \times 0.06$$

$$I = 4.67 \text{ A}$$

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24 Ans: D